

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException
AtTheRateException
DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
import java.util.Scanner;
```

```
// Custom Exceptions
```

```
class DotException extends Exception {  
    public DotException(String message) {  
        super(message);  
    }  
}
```

```
class AtTheRateException extends Exception {  
    public AtTheRateException(String message) {  
        super(message);  
    }  
}
```

```
class DomainException extends Exception {  
    public DomainException(String message) {  
        super(message);  
    }  
}
```

```
public class Main{
```

```
    public static void validateEmail(String email) throws DotException,  
    AtTheRateException, DomainException {
```

```
        // Check for exactly one '@'
```

```
        int atCount = email.length() - email.replace("@", "").length();
```

```
        if (atCount != 1) {
```

```
            throw new AtTheRateException("AtTheRateException: Invalid @ usage");
```

```
        }
```

```
        // Email must not start or end with '.' or '@'
```

```
if (email.startsWith(".") || email.endsWith(".") || email.startsWith("@") ||
email.endsWith("@")) {
    throw new DotException("DotException: Invalid Dot usage");
}

// No consecutive '.' or '@'
if (email.contains("..") || email.contains("@@") || email.contains(".@") ||
email.contains("@.")) {
    throw new DotException("DotException: Invalid Dot usage");
}

// Split local and domain parts
int atIndex = email.indexOf('@');
String localPart = email.substring(0, atIndex);
String domainPart = email.substring(atIndex + 1);

// Check for '.' after '@' (in domain)
if (!domainPart.contains(".")) {
    throw new DotException("DotException: Invalid Dot usage");
}

// Check exactly one '.' after '@' (exactly one dot in domain part)
int dotCountInDomain = domainPart.length() - domainPart.replace(".",
"" ).length();
if (dotCountInDomain != 1) {
    throw new DotException("DotException: Invalid Dot usage");
}

// Get domain extension (part after last dot)
int lastDotIndex = domainPart.lastIndexOf('.');
if (lastDotIndex == domainPart.length() - 1) { // ends with dot
    throw new DotException("DotException: Invalid Dot usage");
}
String domainExtension = domainPart.substring(lastDotIndex + 1);

// Validate domain extension
if (!(domainExtension.equals("in") || domainExtension.equals("com") ||
domainExtension.equals("net") || domainExtension.equals("biz"))) {
    throw new DomainException("DomainException: Invalid Domain");
}
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    String email = sc.nextLine();  
  
    try {  
        validateEmail(email);  
        System.out.println("Valid email address");  
    } catch (DotException e) {  
        System.out.println(e.getMessage());  
        System.out.println("Invalid email address");  
    } catch (AtTheRateException e) {  
        System.out.println(e.getMessage());  
        System.out.println("Invalid email address");  
    } catch (DomainException e) {  
        System.out.println(e.getMessage());  
        System.out.println("Invalid email address");  
    }  
}
```

Status : Correct

Marks : 10/10